# ARGUS: Learning a Bird from Five Calls

Two-Stage Few-Shot Detection of a Threatened Western Ghats Endemic, and a Measurement Gap in How the Field Scores It

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## Table of Contents


Abstract ..... 3
1. Introduction ..... 3
1.1 The monitoring problem ..... 3
1.2 Why few-shot learning ..... 3
1.3 Our research question ..... 4
1.4 Design goals ..... 4
2. Background ..... 4
3. Methodology ..... 5
3.1 Data ..... 5
3.2 Choosing targets and decoys in advance ..... 5
3.3 Stage 1: finding candidate calls ..... 5
3.4 Stage 2: checking the species ..... 5
3.5 Evaluation ..... 5
3.6 Measured noise floor ..... 6
4. Results ..... 6
4.1 Main result: verification helps all ten species ..... 6
4.2 False alarms ..... 7
4.3 The standard test cannot see species mix-ups ..... 7
4.4 Robustness checks ..... 8
4.5 What did not help ..... 8
4.6 Three alternative explanations, tested ..... 9
4.7 Can we remove Perch? ..... 9
4.8 First scan of unlabelled forest audio ..... 10
5. Discussion ..... 10
6. Limitations ..... 10
7. Conclusion and Future Work ..... 11
8. Acknowledgements ..... 11
9. References ..... 12

## Abstract

The Western Ghats is one of Earth's eight hottest biodiversity hotspots and one of its least acoustically monitored. *Sholicola albiventris*, a Western Ghats endemic listed as IUCN Near Threatened, has only 22 labelled recordings in BirdCLEF 2024, far fewer than deep learning normally needs. The species conservationists most need to track are often data-poor for the same reason they are threatened. We asked whether a model can learn a new species from five example calls, and whether it can tell that species apart from birds that sound almost the same.

We built a two-stage pipeline. A small convolutional network, trained from scratch with a prototypical few-shot objective and fine-tuned in seconds for each species, proposes candidate calls. A second stage embeds each candidate with Google's Perch v2 model and scores it with a simple classifier trained on the same five examples. Across 10 Western Ghats species, each tested against its four most similar-sounding species, the first stage alone separated the target from these look-alikes only slightly better than chance (mean AUC 0.585). Adding the second stage improved all 10 species, to a mean AUC of 0.751. The gain is 5.2 times our measured run-to-run noise, held across four independent repeats, and came with four times fewer false alarms on audio containing no target. Three further tests ruled out the obvious alternative explanations: that Perch alone does the work, that the model learned the recording equipment instead of the bird, and a design choice in stage 1.

Because it needs only five calls per species, this approach could help monitor threatened species that have too little data for conventional detectors.

## 1. Introduction

### 1.1 The monitoring problem

The Western Ghats mountain range, along India's western coast, is one of the eight richest biodiversity hotspots on Earth, yet very little of it is acoustically monitored. Passive acoustic monitoring means leaving recorders in the forest and analysing what they capture, and it works in terrain where people cannot survey often or quietly. The hard part is not recording but listening: one recorder collects more audio in a week than a specialist can review in a month.

Automatic detection solves this for common species. The problem is that the species conservationists most need to track usually have the least data. *Sholicola albiventris* (White-bellied Sholakili) lives only in the high-altitude shola forests of the Western Ghats and is listed as Near Threatened on the IUCN Red List [22]. BirdCLEF 2024 has 22 recordings of it, while most of the other species in our study have 500. A standard classifier trained on 22 examples is not realistic: the research group that studies this bird built its own BirdNET detector, and it needed several thousand songs for training (C. Arvind, pers. comm., 4 Sep 2026). Across India, long-term declines are concentrated among habitat specialists and endemics [23], which is exactly the group that is hardest to collect training data for.

### 1.2 Why few-shot learning

Few-shot learning teaches a model a new class from a handful of examples by reusing a general sense of similarity learned on other classes. The DCASE challenge has run a five-shot bioacoustic

detection task every year since 2021 [1, 2, 3], and reported F-scores rose from 38.4% (2021) to 65.2% (2024) [3, 4]. The task is well defined and actively studied.

### 1.3 Our research question

Detecting that a bird called is not the same as identifying which bird called. A system that flags every bird-like sound can score well on a detection benchmark while knowing almost nothing about species. For a conservation record ("this Near Threatened species was at this place on this date"), a confident wrong-species detection is worse than a miss, because it enters the database as a false record.

The DCASE task cannot separate the two. It is scored on one class in one recording: given five example events, find more of the same events in the same file [1]. The system is never asked to tell the target apart from a similar-sounding species. The challenge's own expert reviewers noticed cross-species mistakes in top systems, but the metric has no way to score them [1].

Our question was therefore: can a five-shot detector tell its target species apart from the species that sound most like it, and does adding a separate verification stage measurably improve this?

### 1.4 Design goals

We set measurable goals before running the experiments:

| # | Design goal | How it is measured |
|---|---|---|
| 1 | Learn a new species from five labelled recordings, without retraining the base model | Support set fixed at 5 |
| 2 | Tell the target apart from its four most similar-sounding species | Species-discrimination AUC, target vs. decoy |
| 3 | Beat the first stage alone by more than run-to-run noise | Gain must exceed the measured noise floor |
| 4 | Work beyond the study species | Tested on 10 pre-registered species |
| 5 | Reduce false alarms on audio with no target | False alarms per hour on target-absent audio |

Goal 3 is set against measured noise rather than a fixed number because an early version of this project reported improvements that later turned out to be within run-to-run variation (see Section 6).

## 2. Background

The reference paper for few-shot bioacoustic detection is Nolasco et al. [1], which defines the five-shot task and analyses the 2021 and 2022 challenges. Its baseline, and our first stage, uses prototypical networks [14], which average the five examples into a single point. The paper names this averaging as an open problem, and also reports that the choice of which five examples to use matters a lot. The DCASE 2024 baseline paper [4] adds hard-negative mining and query-time adaptation and compares audio front-ends.

Our second stage follows the transfer-learning approach of Ghani et al. [5]: a large pretrained bird-audio model produces embeddings, and a light classifier is trained on top of them. We use Perch 2.0 [15], which produces 1536-number embeddings, was trained on 14,795 classes, and is designed for

exactly this kind of simple classifier on top. It also transfers well beyond birds, for example to underwater sounds [16].

The failure we measure, a detector that fires correctly but cannot confirm which species it heard, is a form of open-set recognition [19, 21]. We also considered approaches we did not use: self-supervised and general animal-sound encoders [7, 8, 9], refinements of the prototypical baseline [10, 11, 12, 13], and active-learning or agile-modelling workflows in which an expert labels examples over several rounds [20, 26]. Where an expert is available, agile modelling [26] is probably the better tool. Our setting assumes five examples and nothing more.

# 3. Methodology

## 3.1 Data

All audio comes from the BirdCLEF 2024 dataset [6], a public dataset built around under-studied Western Ghats species. The recordings come from Xeno-canto and include the species label, the recordist and a quality rating. Audio is processed at 22,050 Hz mono with the same spectrogram settings as the DCASE 2024 baseline [4], so the pipeline can later be tested on that benchmark without changes.

## 3.2 Choosing targets and decoys in advance

We picked 10 target species with a rule fixed before looking at any results: species with enough recordings, ranked by recording count, with the study species *S. albiventris* forced in first so that it could not be dropped if it did badly. For each target, the four decoys are the species whose Perch embeddings sit closest to it, in other words the hardest available look-alikes. All ten targets are excluded from stage 1's training data, and the code checks this on every run.

## 3.3 Stage 1: finding candidate calls

Stage 1 is a small convolutional network (about 445,000 parameters) trained from scratch on 40 other species with episodic prototypical-network training [14]. For each target it is fine-tuned for 250 steps on the five support calls, and then scans the recording in overlapping one-second windows. Its threshold is deliberately low, because its job is to find possible calls, not to be precise.

## 3.4 Stage 2: checking the species

Each candidate is embedded with Perch v2 [15] and scored by a logistic-regression classifier trained on the same five support calls against calls of other species. Strong regularisation ($C = 0.1$) is needed: with about 100 training points in 1536 dimensions, an unregularised model separates everything perfectly and can no longer rank candidates. For each candidate the pipeline keeps both scores, the stage-1 confidence p and the verifier confidence v.

## 3.5 Evaluation

We plant real target calls into real Western Ghats soundscape recordings, so we know exactly where the answers are. Three tests run on each trained detector:

| Test | What is planted | What it measures |
|---|---|---|
| Detection | Target calls in background audio | How well calls are found; sets the threshold |

| Test | What is planted | What it measures |
|---|---|---|
| Discrimination | Target calls and decoy-species calls | Whether the target is ranked above look-alikes (main result) |
| False alarms | Nothing | False alarms per hour (every detection is wrong) |

Test calls never overlap with the five support calls, and calls are planted at +6 dB signal-to-noise ratio unless stated otherwise. The main experiment uses three training seeds per species. The robustness sweeps use 5 species × 3 seeds, which gives 15 measurements per condition.

Our main measure is species-discrimination AUC: if we take one target call and one decoy call at random, how often does the system score the target higher? 0.5 is guessing and 1.0 is perfect. It needs no threshold, which is also why BirdCLEF 2024 chose ROC-AUC as its official metric [6]. Precision-recall measures are often preferred for rare events [18], but our discrimination test is only mildly imbalanced (6 targets against 12 decoy calls), so AUC is appropriate. We also report average precision (AP) and F1, because the difference between them and AUC turned out to be one of our findings (Section 4.3).

### 3.6 Measured noise floor

GPU computations are not perfectly repeatable, so identical runs give slightly different results. We measured this directly: running the same configuration with three seeds on three species gave spreads of 0.035, 0.029 and 0.032 AUC, a mean noise floor of 0.032. We only count a difference as real if it is larger than this.

All results derive from eleven CSV files in the data folder of our repository, and every number in this paper was recomputed from them while writing. The pipeline's logic is covered by local tests that run without a GPU.

## 4. Results

### 4.1 Main result: verification helps all ten species

Across 10 species and 3 training seeds:

| | Stage 1 alone | With verification |
|---|---|---|
| Mean species-discrimination AUC | 0.585 | 0.751 |
| Mean gain | | +0.165 |
| Species improved | | 10 / 10 |
| Gain ÷ measured noise floor (0.032) | | 5.2× |

Per species, sorted by gain:

| Species | Stage 1 | Verified | Gain |
|---|---|---|---|
| comgre | 0.687 | 0.781 | +0.094 |
| comsan | 0.623 | 0.720 | +0.097 |
| blrwar1 | 0.535 | 0.646 | +0.111 |
| commoo3 | 0.628 | 0.750 | +0.122 |
| bkwsti | 0.694 | 0.828 | +0.134 |
| comkin1 | 0.621 | 0.827 | +0.205 |

| Species | Stage 1 | Verified | Gain |
|---|---|---|---|
| eaywag1 | 0.545 | 0.753 | +0.208 |
| whbsho3 (study species) | 0.469 | 0.680 | +0.211 |
| barswa | 0.554 | 0.770 | +0.216 |
| bcnher | 0.496 | 0.750 | +0.254 |

Every species improves, and even the smallest gain (+0.094) is about three times the noise floor. For two species, our study species (0.469) and bcnher (0.496), stage 1 alone is at or below chance, so all of the species information comes from verification. We ran the whole experiment in four independent sessions, and the mean gains were +0.165, +0.166, +0.161 and +0.165.

### 4.2 False alarms

On soundscapes with no planted target, at the best-F1 threshold:

| | Stage 1 alone | With verification |
|---|---|---|
| Mean false alarms per hour | 54.0 | 13.4 |
| Reduction | | 4.03× |
| Worst species | 78.8 (whbsho3) | 54.2 (comkin1) |
| Best species | 23.0 (bkwsti) | 0.7 (barswa) |

These soundscapes are real forest audio, so any wild target call already in them is counted as a false alarm, and the true rates can only be lower. The comparison holds at the best-F1 threshold only. At a 90%-recall threshold it cannot be made, because recall never reaches 0.9 (the peak is 0.778).

### 4.3 The standard test cannot see species mix-ups

| Measure | Stage 1 | With verification | Change | Species improved |
|---|---|---|---|---|
| Detection average precision (AP) | 0.611 | 0.604 | −0.008 | 4 / 10 |
| Event F1 | 0.675 | 0.710 | +0.035 | 9 / 10 |
| Species-discrimination AUC | 0.585 | 0.751 | +0.165 | 10 / 10 |

We also compared six ways of combining the two scores:

| Fusion rule | AUC | AP |
|---|---|---|
| v alone (verifier only) | 0.751 | 0.604 |
| sqrt(p) · v | 0.650 | 0.636 |
| p · v | 0.626 | 0.625 |
| sqrt(p · v) | 0.626 | 0.625 |
| p · sqrt(v) | 0.609 | 0.619 |
| p alone (stage 1 only) | 0.585 | 0.611 |

Ranked by AP, the standard detection measure, the verifier-only rule comes last. Ranked by discrimination AUC, it comes first. AP only asks whether calls are found against background noise. It never checks whether the system can tell the target from a similar species, so anyone choosing a configuration by AP alone would discard the one that identifies species best. Using the verifier's score on its own also beat every blend: stage 1's confidence is useful for finding candidates, but adds no species information once the verifier has scored them.

### 4.4 Robustness checks

**Number of examples.** Verified AUC was 0.680 with one example, 0.720 with three, 0.726 with five and 0.758 with ten (5 species × 3 seeds). Only the step from one to three clearly exceeds the noise floor, so five is a reasonable point on a smooth curve rather than a sharp optimum. Stage 1 alone stayed between 0.543 and 0.568 at every count: extra examples improve the verifier, not stage 1.

**Which five examples.** Choosing the five at random, by quality rating, for diversity, or simply taking the first five gave verified AUCs between 0.715 and 0.748. That spread (0.033) is the same size as the noise floor, so no strategy reliably wins. Taking the first five, which the DCASE protocol requires, came last in both of our samplings, but not by a significant margin.

**Call loudness.** We varied the loudness of planted calls from +9 dB down to −6 dB. Coverage, the share of target calls the system finds at all, fell from 0.813 to 0.647, and AUC over all planted calls fell from 0.737 to 0.634. Over the calls it did find, however, AUC stayed almost flat (0.848 to 0.820). In quiet audio the system misses more calls, but when it finds one it identifies the species just as well. This is why we report coverage next to AUC.

**Decoy difficulty.** Easier decoys raise absolute AUC, as they must, but verification's gain barely changes: +0.180 against the hardest decoys, +0.162 against medium ones and +0.176 against random ones. That spread (0.018) is inside the noise floor. Easier decoys would have given us a higher headline number, so the hardest-decoy result we report is the conservative one.

### 4.5 What did not help

We tested published techniques in a factorial design (8 configurations, 3 species): a prototypical-probe verifier from the Perch 2.0 benchmark [15], hard-negative mining [4], and a log-Mel front-end of the kind used by the DCASE 2024 winning entry and the DCASE 2024 baseline paper [4].

| Configuration | Verified AUC | False alarms per hour |
|---|---|---|
| pcen, logreg (our baseline) | 0.742 | 8.2 |
| pcen, proto | 0.724 | 27.8 |
| pcen, logreg + hard-neg | 0.676 | 12.3 |
| pcen, proto + hard-neg | 0.667 | 37.3 |
| logmel, logreg + hard-neg | 0.645 | 8.0 |
| logmel, logreg | 0.605 | 3.3 |
| logmel, proto + hard-neg | 0.598 | 28.8 |
| logmel, proto | 0.588 | 37.0 |

The untouched baseline beat every alternative. The prototypical probe and hard-negative mining raised false alarms by 3× to 19× across runs. Log-Mel appears to have the fewest false alarms (3.3 per hour), but only because it detects far less: verified recall fell from 0.610 to 0.366, and stage-1 recall from 0.570 to 0.359. A fourth technique, the TimeFilterAug augmentation from the DCASE 2023 winning entry [17], made every metric worse in earlier tests and was dropped. The published gains for these methods come from fully trained systems. In our five-shot setting, with a 250-step fine-tune, they did not transfer.

### 4.6 Three alternative explanations, tested

An outside review raised three ways our result could be misleading. We tested all three on 5 species (data/argus_confounds.csv):

| Question | Test | Result |
|---|---|---|
| Is Perch doing all the work? | Run Perch alone as the detector, with no stage 1 | Perch alone: AUC 0.526 and only 2.5% recall. Full pipeline: 0.735. Neither stage works without the other |
| Did stage 1's negative examples drive the result? | Replace background negatives with real calls of other species (6 seeds) | Verified AUC fell from 0.737 to 0.686, lower in 5 of 5 species. Harder negatives made it worse |
| Did the model learn the recordist's equipment instead of the bird? | Test only on recordings by different recordists from the support calls | Verified AUC 0.740 to 0.708; the change (−0.032) is inside its noise bound of 0.062 |

The second result clears its threshold only narrowly. What makes it believable is that it goes the same way in all five species and agrees with the hard-negative result in Section 4.5.

### 4.7 Can we remove Perch?

Perch is a large model (about 12 million parameters) that cannot run on a battery-powered field recorder, and it is not something we built. We therefore tried to distil Perch's embeddings into our small stage-1 network, so that no Perch is needed at detection time. A small head was trained to reproduce Perch's embedding of the same audio, using 2,636 examples from 60 other species, with 9 species held out to test generalisation. The distilled model has 642,688 parameters, about 19 times smaller than Perch.

| Species | Perch AUC | Distilled AUC | Change |
|---|---|---|---|
| bkwsti | 0.814 | 0.803 | −0.011 |
| barswa | 0.789 | 0.718 | −0.071 |
| blrwar1 | 0.594 | 0.537 | −0.056 |
| whbsho3 (study species) | 0.672 | 0.511 | −0.161 |
| bcnher | 0.795 | 0.412 | −0.383 |

On average, discrimination fell by 0.136 AUC and false alarms rose from 10.2 to 84.2 per hour. For one species the loss is almost nothing, but our study species dropped to chance (0.511). We report this as a negative result.

When we asked about the uneven losses, the author of Perch suggested two causes: too little or unbalanced training data for some species, and too little model width (T. Denton, pers. comm., 4 Sep 2026). We tested width directly by widening the student eightfold (64 to 512 dimensions), trained on a larger set of 10,163 examples from 98 species. Similarity to Perch rose steadily (0.4446 to 0.5226), but species AUC did not move (0.610 to 0.611; the total spread across widths was 0.0073, well inside the noise floor). A student can get closer and closer to Perch's embeddings without getting any better at telling similar birds apart. The larger training set did lift the study species from 0.511 to 0.608, but other settings changed at the same time, so this is suggestive rather than controlled. For now, the pipeline still needs Perch.

### 4.8 First scan of unlabelled forest audio

Everything above uses calls we planted ourselves. As a first real-world check, we scanned 400 unlabelled soundscape recordings (26.6 hours, kept separate from all evaluation audio) for the study species and for a control species with a very different habitat, the Black-winged Stilt (*Himantopus himantopus*), on exactly the same files.

| | **Detections** | **Files** | **Rate** |
|---|---|---|---|
| *S. albiventris* | 350 | 55 | 13.2 per hour |
| Control (*H. himantopus*) | 201 | 66 | 7.6 per hour |

On identical audio, the study species was flagged 1.74 times as often as the control. This is the first sign that the pipeline responds to which species it was given, rather than to any bird-like sound. It does not confirm any detection. There is no ground truth for these files, nobody on our team can identify this bird by ear, and every score sits only just above its threshold (the highest is 0.526 against a threshold of 0.491). We saved the 60 highest-scoring clips for each species so that an expert can review them, which has not happened yet.

## 5. Discussion

The results show a clear division of labour. Stage 1 is a locator: fast and good at finding calls, but close to chance at telling species apart, however many examples it is given. Stage 2 is an identifier: good at species, but nearly useless at finding calls on its own (2.5% recall when Perch scans by itself). Each stage fails at the other's job, which answers both "it is all Perch" and "then why not just use the CNN".

The gap between detection scores and discrimination scores matters beyond this project. A configuration chosen by soundscape AP alone can be the worst one at separating a species from its look-alikes. For few-shot systems that are meant to produce species records, we suggest reporting a discrimination measure against similar-sounding species alongside the usual retrieval scores.

A system like this has two different uses. As a labelling aid, where a person reviews every candidate, recall matters most and false alarms are cheap, so stage 1 alone (54 false alarms per hour) is fine. As a record that a threatened species was present, a wrong-species detection does real harm, and the verified pipeline (13.4 false alarms per hour) is the right choice. Our work is aimed at the second use.

## 6. Limitations

1. **No external benchmark.** All comparisons are within our own pipeline. Running the DCASE Task 5 development set [1, 4] is the biggest missing piece; our front-end already matches its baseline. None of our numbers should be compared with published DCASE F-scores.
2. **No confirmed wild detection.** The background audio is real, but the target calls are planted. The wild scan (Section 4.8) has been attempted but not yet checked by an expert.
3. **The five calls may come from only a few birds.** This species shows cultural and genetic variation across its range [24], and a researcher who studies its song told us that variation between individuals is very high (C. Arvind, pers. comm., 4 Sep 2026). Twelve of the 22 recordings come from a single team at a single location, so a random set of five has a 59% chance of including at least three from that team. Our results may describe a few individual

birds better than the species as a whole. Choosing the five calls across different recordists is the next experiment, and a published method for measuring song complexity in this species [25] could help.

4. **The pipeline still needs Perch** at detection time (Section 4.7). The best distilled version still trails Perch by 0.094 AUC.
5. **Modest sample sizes.** The main experiment uses 3 seeds on 10 species, and the sweeps use 15 measurements per condition. Ten seeds per condition would be better; our limit was free GPU time.
6. **The study species has the least data.** It has 22 recordings, against 500 for most of the others, so its result (0.469 to 0.680) rests on the least evidence of the ten.

During the project we also withdrew seven of our own earlier claims after re-checking them. For example, we had claimed that performance plateaus at five examples and that one way of choosing examples was best; neither survived a properly powered test (Section 4.4). We also found and fixed four bugs before reporting any results, including one that let some target species leak into stage 1's training data. The full dated record is kept in our lab notebook.

## 7. Conclusion and Future Work

A bioacoustic detector can learn a new species from five calls and tell it apart from its closest look-alikes, but only with a verification stage. Stage 1 alone reaches a mean AUC of 0.585. Adding Perch-based verification lifts all ten species to 0.751, 5.2 times our measured noise floor, repeated in four independent runs, with four times fewer false alarms. The standard detection measure (AP) barely moves (0.611 to 0.604) and would rank our best configuration last, which shows that detection scores alone can hide species mix-ups. Four published techniques did not help in this five-shot setting, and compressing Perch into our small network does not yet work for the study species.

Future work, in order of priority:

1. Validate on the external DCASE Task 5 benchmark.
2. Measure how results depend on how many different recordists, or individual birds, the five calls come from.
3. Design a distillation objective that preserves species separation rather than embedding similarity.
4. Increase the number of seeds for the results that sit close to the noise floor.
5. Run the system on long field recordings with expert-confirmed labels.
6. Turn the discrimination test into an evaluation protocol that other projects can reuse.

## 8. Acknowledgements

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distillation failed unevenly across species. Neither of them reviewed this paper, and any errors are ours.

This work used the public BirdCLEF 2024 dataset [6] and the public Perch v2 model [15], with computation on a free GPU platform. Generative AI tools (Claude, Antigravity) wrote much of the code and helped draft the text of this paper. All experiments and results are our own.

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